

# ROKIB AL DHIN RAADH

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*Independent AI researcher and founder specializing in autonomous, self-correcting neural architectures and multi-agent systems.*

## Publications & Technical Reports

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**R.A.D. Raadh.** “ORMAS: Neural Architectural Transparency Enables Autonomous Self-Correction.” 2026. Zenodo preprint. DOI: [10.5281/zenodo.21730363](https://doi.org/10.5281/zenodo.21730363).

Submitted to **AAAI 2027**, Main Technical Track — under anonymous review. Not accepted and not published.

## Academic Service & Recognition

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| Dec 2026     | <b>Reviewer, Program Committee</b><br>Trustworthy AI for Good (AI4GOOD) Workshop — <b>NeurIPS 2026</b> , Paris <ul style="list-style-type: none"><li>• Peer review for the workshop’s trustworthy-AI track, covering mechanistic interpretability, attribution, post-deployment monitoring and model auditing.</li><li>• The ICML 2026 edition of the workshop drew 539 submissions against a 34% acceptance rate, assessed by a committee of 237 reviewers.</li></ul> |
| Jul 2026     | <b>Cosmos Institute</b> — Grants Review<br>Application ranked <b>highest in its cycle</b> . Not funded: that round backed philosophical work on AI, and ORMAS is a technical architecture. Invited to reapply to a technical track.                                                                                                                                                                                                                                    |
| Aug–Sep 2026 | <b>Entrepreneur First</b> — The Bridge, San Francisco<br>Cold application answered within <b>12 hours</b> with a call booked with an EF investment-team lead; first screen held in London, August 2026. Now in the selection process for <b>The Bridge</b> , EF’s eight-week San Francisco residency — two interview rounds completed, decision pending as of September 2026.                                                                                          |

## Research Experience

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2024—pres.

## Independent Machine Learning Researcher

ORMAS: Neural Architectural Transparency Enables Autonomous Self-Correction

PyTorch · 10,594 lines · 61 files · Codebase: [anonymous.4open.science/r/ormas-EB73](https://anonymous.4open.science/r/ormas-EB73)

- **Theory & Architecture:** Designed a three-signal training architecture: global backpropagation, a per-node local loss anchor (4,715-param bottleneck incapable of memorising noise), and a health-gated correction engine diagnosing 8 pathology types with expert immunity and selective rollback.
- Proved the first Input-to-State Stability (ISS) convergence guarantee for any self-correcting neural architecture; empirically validated by 83–93% correction magnitude decay over 200 epochs.
- **Catastrophic Shock Recovery:** Engineered ORMAS to autonomously recover to **80.3%** accuracy after its primary feature layer was zeroed mid-training — reclaiming 97% of lost performance. Standard CNNs collapse permanently to 10.0% (zero variance,  $n=3$ ).
- Demonstrated recovery to **72.9%** (+62.9 pp gap) after all three convolutional layers were simultaneously zeroed — a scenario causing permanent, unrecoverable collapse in standard architectures.
- Under extreme weight perturbation ( $\sigma=2.0$ ), ORMAS recovered to **62.1%** vs. 38.2% for a standard CNN (+52.1 pp) — the recovery gap widens monotonically with perturbation scale.
- **Generalization & Robustness:** Achieved **94.6%** shape memory retention during sequential task learning (ResNet-18: 47.3%); produced 58.8% zero-shot accuracy on novel unseen shape+colour combinations vs. 25% chance.
- Validated noise robustness across 6 regimes: asymmetric 40% corruption 80.6%; CIFAR-10N real annotator noise  $1.4\times$  more stable than standard training;  $<0.8\%$  accuracy variance across a  $16\times$  hyperparameter range.
- **Scale & Reproducibility:** 383 controlled experiments across 4 architectures; 340 million telemetry entries, every correction causally tagged to pathology, node, and epoch — single RTX 3090.

## Engineering

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2023—pres.

## Sole Architect & Engineer

OXIMO — Cognitive Multi-Agent Operating System

Python · 40,933 lines production · 2,011 tests · Codebase: [anonymous.4open.science/r/oximo-5C73](https://anonymous.4open.science/r/oximo-5C73)

- Built autonomous multi-agent system with hierarchical task decomposition, on-demand role creation via self-hiring pipeline, Graph-of-Thoughts planner, and 3-tier cognitive memory with temporal decay.
- Rebuilt from a **106,000-line monolith** — **72% line reduction** while preserving all **12 critical algorithms** with full audit trail. 47 feature flags across 17 implementation phases.
- Defense-in-depth safety: 7-check input cascade, PII redaction, injection pattern matching, and behavioral drift monitoring via EMA with Page-Hinkley change detection.

## Ventures & Deployment

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| 2024–pres. | <p><b>Founder</b>    <i>Black Bloxie LTD — UK Registered Holding Company</i></p> <p>A private equity structure deploying OXIMO as a scalable replacement for human operations across 4 verticals.</p> <ul style="list-style-type: none"><li>• <b>+1,300% revenue recovery &amp; 99% cost reduction:</b> Replaced human-operated workflows and expensive monolithic API calls (e.g., GPT-4) with an orchestrated cascade of six specialized, smaller AI models. Achieved equivalent output quality at 1% of the cost.</li><li>• <b>Vertical 1 (E-Commerce) — Causal Proof:</b> 3-phase ablation over 1.5–2 years. Phase 1 (OXIMO deployed): 500+ customers, \$0 ad spend. Phase 2 (AI removed): revenue –91%, new customers –100%. Phase 3 (V3 re-injected): revenue +1,300%. Proved the system, not the market, drives the revenue.</li><li>• <b>Emergent AI-to-AI economy:</b> 100% of acquisition driven autonomously by LLM referral traffic (ChatGPT, Bing, Gemini). Paid ads: zero. Human sales effort: zero.</li><li>• <b>Verticals 2–4 (Law, Software, Medical Research):</b> Actively running. Full autonomous hiring, dynamic pricing, and self-modification validated by 2,011 tests; static boundaries currently held pending AI accountability regulation.</li><li>• Registered at 17 from Bangladesh; navigated UK company formation, AML compliance, and cross-border operations independently.</li></ul> |
| 2020–2024  | <p><b>Independent Ventures (5)</b>    <i>Automated Sales · AI Development · Market Tools</i></p> <ul style="list-style-type: none"><li>• Iterative testing across 5 independent ventures drove the architectural constraints and R&amp;D foundation for OXIMO.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Technical Skills

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| ML / AI     | PyTorch, PCGrad gradient surgery, ISS convergence theory, noise-robust training, multi-agent orchestration |
| Engineering | Python (advanced), C++, LaTeX, Docker, FastAPI, SQLAlchemy, ChromaDB, Prometheus, React, TypeScript        |
| Languages   | English (fluent), Bengali (native)                                                                         |

## Education

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| Fall 2027 | <p><b>Prospective Undergraduate</b></p> <p>Seeking admission to pursue research in AI safety, self-correcting neural architectures, and autonomous systems.</p>                                                                                                                                                                               |
| 2025–2026 | <p><b>Self-Directed Coursework</b></p> <ul style="list-style-type: none"><li>• Deep Learning Specialization (5 courses), DeepLearning.AI (Andrew Ng). <i>Completed.</i></li><li>• MITx AI Programs — 25 courses across AI foundations, precision medicine, sustainable energy, and transportation. <i>Certificates in progress.</i></li></ul> |

## Open-Source & Archives

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| ORMAS    | <a href="https://zenodo.org/records/21730363">zenodo.org/records/21730363</a> (permanently archived)                |
| OXIMO    | <a href="https://anonymous.4open.science/r/oximo-5C73">anonymous.4open.science/r/oximo-5C73</a> (anonymous review)  |
| Codebase | <a href="https://anonymous.4open.science/r/ormas-EB73">anonymous.4open.science/r/ormas-EB73</a> (ORMAS source code) |
| Data     | <a href="https://drive.google.com/">drive.google.com/...</a> (383 experiments)                                      |